

Lent Week 2 Report: Synthetic Datasets

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1 Completed Tasks

This week, the following tasks were completed:

- Implemented and generated six synthetic datasets with different structures to evaluate Additive and OAK kernels.
- Hypothesized expected model behavior for each dataset.
- Conducted experiments to compare the performance of Additive and OAK kernels.
- Generated Sobol indices for each dataset and kernel type.
- Visualized the results, including performance metrics and feature contributions.

2 To discuss

- TMR
- Industry Feedback
- Aims and action points
- Model capacity $|k(\mathbf{x}, \mathbf{x}')|^{-1/2}$, if same then is it just a optimisation issue? e.g. if i optimise on one and initialise with those hyperparams on the other will it give same performance?

2.1 David Duvenaud Email:

I also looked into your 'pathologies in very deep networks' paper with the hope that it might provide insight into OAK's performance improvements. As you mentioned, neural networks with dropout regularisation can be reinterpreted in the Gaussian Process sense, and I am hoping this reinterpretation may hold some clues as to why we see the performance change with OAK. I was wondering if you may have intuition or thoughts on comparing the regularisation provided by dropout and by the orthogonality constraint in the OAK paper. I suspect that in traditional additive GPs, the lack of orthogonality might lead to redundancy and inefficiency in modelling complex functions, which is having subsequent overfitting effects and leading to a relative reduction in performance.

Testing these thoughts

- To test this hypothesis, I am considering the following experimental setup:
- Use synthetic noisy data with a simple additive structure (e.g., $y=f_1(x_1)+f_2(x_2)+\text{noise}$).
- Train and compare:
- A traditional additive kernel model (without explicit regularisation).
- An OAK model with orthogonality constraints.
- A traditional additive kernel model with explicit regularisation (e.g., L2 regularisation on kernel parameters).
- Vary the noise levels.

- Evaluate metrics such as the generalisation performance and the sensitivity of model parameters to noise.
- If my intuition is accurate (of which there is no guarantee!) then I believe OAK should generalise better under noise compared to traditional additive models, even when explicit regularisation is applied to the latter.

3 Synthetic Dataset Overview

The synthetic datasets were designed to test various properties of the Additive and OAK kernels. Table 1 summarizes the datasets and their characteristics.

Dataset	Features	True Function	Expected Kernel Behavior
Type 1 (Purely Additive)	3	$y = \sin(x_1) + \log(1 + x_2) + \exp(x_3)$	Both should perform well
Type 2 (Strong Interaction)	2	$y = x_1 \cdot x_2$	OAK should perform better
Type 3 (Sparse Dependency)	3	$y = \sin(x_1) + \log(1 + x_3)$	OAK should ignore x_2
Type 4 (Low-Dimensional Interaction)	3	$y = \sin(x_1) \cos(x_2) + \exp(-x_3)$	OAK should capture interactions
Type 5 (Mixed Additive and Interaction)	4	$y = x_1^2 + x_2^2 + x_3x_4$	OAK should separate additive vs interaction
Type 6 (High-Dimensional Redundancy)	10	$y = \sin(x_1) + x_2^2 + 0.1 \sum_{i=3}^{10} x_i$	OAK should filter out noise

Table 1: Summary of Synthetic Datasets and Expected Kernel Performance

4 Experimental Results

Table 2 presents the average RMSE and NLL results for each dataset.

Dataset	RMSE		NLL	
	Additive	OAK	Additive	OAK
Type 1	0.000187	0.000266	-5.9165	-5.4937
Type 2	0.000028	0.006578	-5.9704	-5.1648
Type 3	0.000068	0.000203	-5.9550	-5.8552
Type 4	0.000080	0.005002	-5.8628	-4.3179
Type 5	0.000308	0.031872	-5.9396	92.2438
Type 6	0.000122	0.020308	-5.9358	1739.4694

Table 2: Averaged RMSE and NLL Performance for Additive and OAK Kernels

5 Observations and Discussion

- **Additive vs OAK Performance:** In some cases (Type 5 and Type 6), OAK produced extreme errors.
- **Feature Selection:** OAK doesn't seem to perform better even in higher dimensional settings. Could be a tuning issue?
- **Sobol Indices:** OAK's Sobol indices provided a clearer separation between additive and interaction effects.
- **Unexpected Findings:** Apart from the performance being one-sided, the performance of OAK on Type 5 and Type 6 showed extreme NLL values, indicating potential numerical instability in high-dimensional settings.

6 Future Work

- **Check optimisation**
- **Analyze convergence behavior** of the kernels on real-world datasets.
- **Investigate why OAK fails** on certain high-dimensional cases.
- **Experiment with different kernel parameterizations** for better interpretability.
- **Automate result extraction** from JSON files for faster report generation.

7 Figures





